

Table 2: SYN scan client responses

Port status	Client response	Inference
Open	Standard response SYN-ACK	Service running on the port
Closed	Standard response RST	Service not running on the port
Filtered	No response	Firewalled port

Table 3: Summary of SYN, ping and UDP scans

Scan type	Facets
SYN scan -sS Scan TCP ports	- Does not leave a log entry - Requires root access. - Traffic of RST frames increases with use of SYN scan. - Gives information about TCP ports.
Ping scan -sP Identify active hosts	- Very difficult to trace – only two standard ICMP frames, which are very common in network traffic, are required to complete the scan. - Root privilege not required to run the scan. - Yields a device inventory by identifying active devices on the network.
UDP scan -sU Scan UDP ports	- Uses 0 byte UDP data, causing low overhead on the network. - Requires root access. - Many operating systems put restrictions on UDP traffic, thus this scan can be very slow if run on devices running those operating systems - Works well on Microsoft operating systems, since Microsoft does not restrict UDP port traffic. - Best for scanning known UDP ports used by spyware/malware for communication.

If the device responds with a SYN-ACK, NMap sends an RST instead of an ACK, resetting the session, rather than completing the handshake for data transfer. If ACK was sent instead of RST, the connection would be left open till session time-out, making the device prone to a denial of service type of situation.

To run a SYN scan, root privileges are required under Linux.

A SYN scan is used to find the status of TCP ports on various devices on the network. Since the SYN scan works on TCP, it will work across all operating systems and other devices that implement TCP, such as controllers, PLCs, network printers, Ethernet switches, and mobile phones. Since it does not open a valid TCP connection, it's quiet, and difficult to detect. However, careful network monitoring will reveal too many RST frames in traffic, due to one RST frame per scanned port. Here's a sample

SYN scan that will return various open TCP ports:

```
nmap -sS 192.168.100.100
```

Ping Scan -sP

This scan is used to find active hosts in the range. Rather than using ports like a SYN scan, a ping scan starts by sending an ICMP echo request to the target range. Active devices on the network will respond with an ICMP echo reply, thus revealing their status. A firewalled host with blocked ICMP will not respond to the ICMP echo request. The obvious basic use of this scan is to find all active hosts on the network. This set of two commands gives a list of all active IP addresses in the 192.168.100.0/24 range:

```
nmap -sP -n -oG hostlist 192.168.100.0/24 (Grep'able output file - hostlist)
cut -d ' ' -f2 hostlist > iplist (List of all active IPs in the target range - iplist)
```

The ping scan uses only one packet for the request, and may get one packet in response, thus making it the fastest of all NMap scan types, with the lowest footprint. The ping scan cannot be combined with other scan types.

UDP Scan -sU

This is used to find the status of UDP ports in the target range. At the start of the UDP scan, NMap sends a 0-byte UDP packet directed towards a UDP port. The target's response differs depending on the status of the scanned port:

- Open port: Data on the scanned UDP port.
- Closed port: ICMP error message indicating no service is running on this port.
- Open/Filtered port: No ICMP message; NMap waits for the time-out, and can't determine whether the port is open, or filtered by a firewall.

UDP can be used to detect malware/spyware effectively. The following sample UDP scan command will return open/closed/open/filtered UDP ports on the host:

```
nmap -sU 192.168.100.100
```

Please try out these scanning techniques, hands-on, before further exploring various other scan options provided by NMap. And don't forget to keep a watch on this column for further details! 

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